Results: Scholarship Coverage and Evidence-Status Audits

Citation coverage, locator health, and evidence-class auditing I

The manuscript's source coverage is also treated as a generated result. Every authored citation is parsed from the section files, matched against `references.bib`, and mapped through `data/scholarship_sources.yaml` into a source lane and support role before the build writes figures. The current audit covers 188 cited sources; that number is generated from the manuscript and source matrix, not maintained as a prose claim. fig. 1 shows both sides of the contract: the left panel compares cited sources with the matrixed source pool for each lane, while the right panel shows which manuscript sections carry each lane. The audit is deliberately orthogonal to the numeric claim registry: one gate protects computed quantities; the other protects the scholarly support surface.

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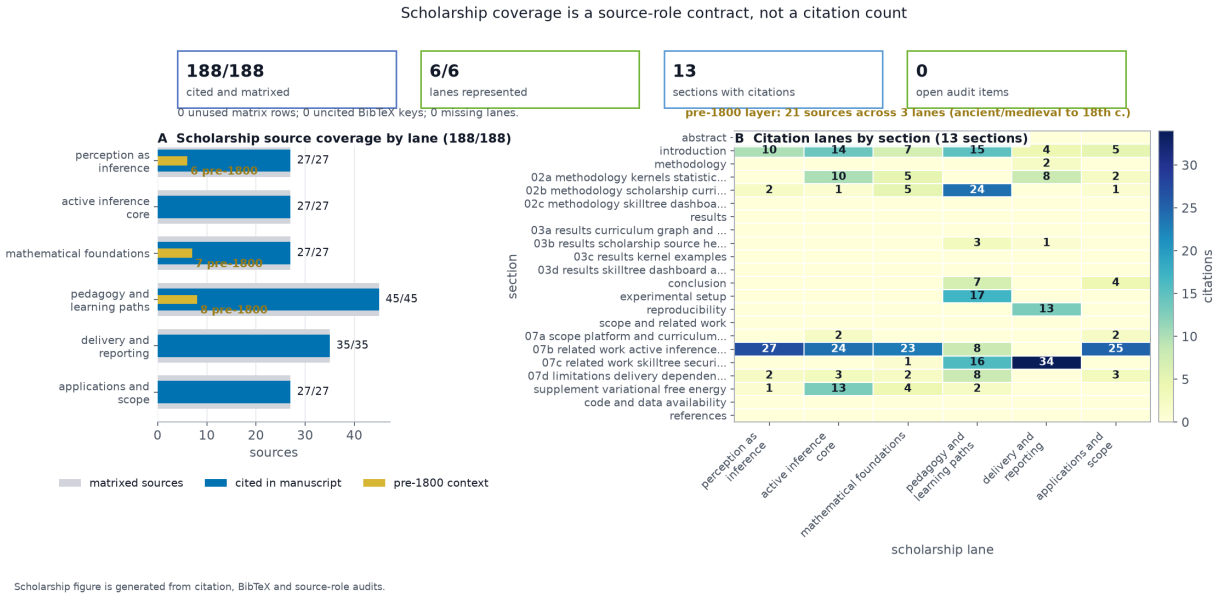


Figure 1: Scholarship coverage audit generated from manuscript citations, BibTeX keys, and the source-role matrix. The left panel compares cited versus matrixed sources by scholarship lane; the right panel maps manuscript sections to citation lanes. This figure makes support distribution inspectable: a large bibliography is not treated as sufficient unless sources are cited in the sections and roles where they support specific claims.

The open-science education layer is also explicit. Active Skillference cites the Active Inference Institute's public education ecosystem, including the open-source course infrastructure, the START curriculum-generation project, and the Institute website's open-science educational mission (Active Inference Institute 2026b, 2026d, 2026a). The generated local Learn material also links every exported skill to commit-pinned external material pointers, including the Active Inference Institute cognitive knowledge base (Active Inference Institute 2026c). fig. 2 shows both surfaces together: cited public sources remain visible, and the course-material source rows expose how the cognitive knowledge base enters the generated skill records.

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Figure 2: Active Inference Institute open-science learning sources across scholarship support and generated course-material pointers. Summary cards report local material records, external pointers, cognitive knowledge-base pointers, and topic-matched KB skills; ledger rows distinguish the course repository, cognitive knowledge-base source, Institute homepage, website repository, and START source with locator and role text. The website repository is a community/resource-hub pointer rather than a course or knowledge-base record. The figure records pinned links and cited support surfaces; it does not claim endorsement, completeness, or copied ownership of the public education ecosystem.

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The source matrix is then checked a second way: source locators must be syntactically specific, documentation entries must use URL locators, and each support role must be substantive enough to explain why the source is present. fig. 3 reports the resulting offline source-health audit and provides a place for optional online DOI/URL reachability checks to surface when release review requires currentness evidence.

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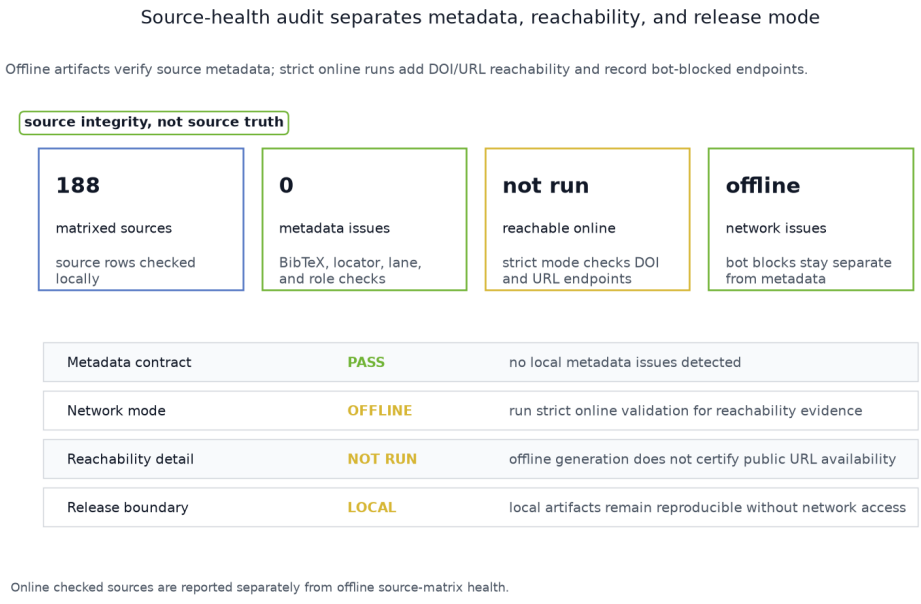


Figure 3: Scholarship source-health audit generated from locator syntax, lane assignment, and support-role checks. Status cards separate matrixed source count, offline validation issues, and online-check mode, while the ledger rows distinguish local metadata health from optional network reachability. The figure separates two different claims: a source can be present in the matrix and still require locator, role, or currentness repair before release.

The validation surface is intentionally visible rather than left as a hidden CI detail. fig. 4 summarizes which gates bind generated JSON, negative controls, manuscript text, dashboard artifacts, release documentation, and SkillTree handoffs. Its count panel is rebuilt from the same graph, claim registry, scholarship audit and roadmap state that produce the curriculum and manuscript variables. The result is a falsifiable release claim: if a count, citation, figure, dashboard artifact, release inventory row, or roadmap item drifts away from its generator, the validator fails before the manuscript or SkillTree export can be presented as current.

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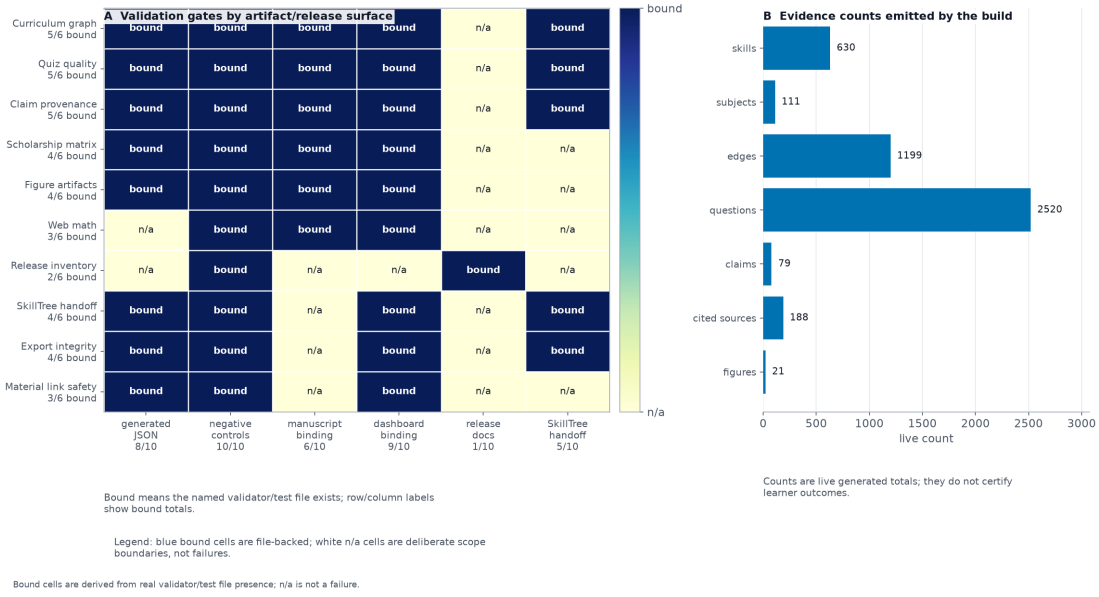


Figure 4: Validation-surface matrix for Active Skillference. Each cell is derived from the on-disk presence of the validator or test file that binds that artifact surface: a cell reads "bound" only when every file named in its binding exists, so deleting a gate flips its cell to "n/a" rather than leaving a stale "bound" claim. Row and column labels state bound-cell totals, and each cell carries text, so the figure remains readable without heatmap color. The side panel reports live evidence counts for skills, subjects, dependencies, assessment questions, claims, cited sources, and registered figures, read directly from the graph, claim registry, scholarship audit, and figure inventory. The interpretation boundary is important: a bound cell means a named validator file exists for that artifact surface (not that the gate is guaranteed sound), while an unbound cell marks a surface that is out of scope for that gate.

fig. 5 makes the negative controls behind that release view explicit. It separates missing registry rows, missing files, missing manuscript labels, missing dashboard artifacts, missing roadmap metrics, and stale manifests so the validation layer is judged by concrete false-certification classes rather than by the presence of a polished figure alone.

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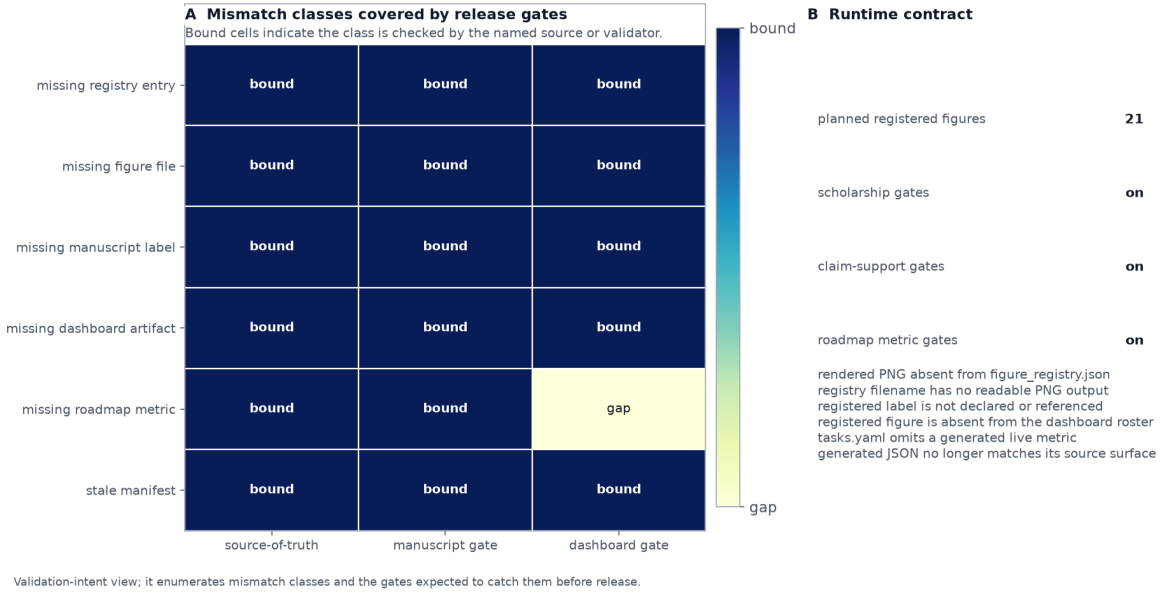


Figure 5: Validation-intent mismatch matrix for Active Skillference. Rows enumerate concrete false-certification classes: a rendered PNG omitted from the registry, a registered figure without a readable file, a registered label omitted from the manuscript, a registered figure omitted from the dashboard artifact roster, a missing roadmap live metric, and a stale generated manifest. Columns show whether the source-of-truth, manuscript gate, or dashboard gate is expected to reject each mismatch before release. The right panel records the planned registered-figure count and which optional audit surfaces were enabled for the render.

The same contract is now represented inside the curriculum rather than only enforced at release time. The visual-evidence slice adds a prerequisite-gated authoring path for question contracts, data grain, encoding strength, annotation boundaries, static/mobile readability, and negative-control validation. fig. 6 shows that path as generated curriculum evidence: each rung names the live skill that carries the review step, while the registry and dashboard expose the same structured chart-contract fields. This is a deterministic artifact-integrity claim, not a claim that the visual curriculum has been externally tested with learners.

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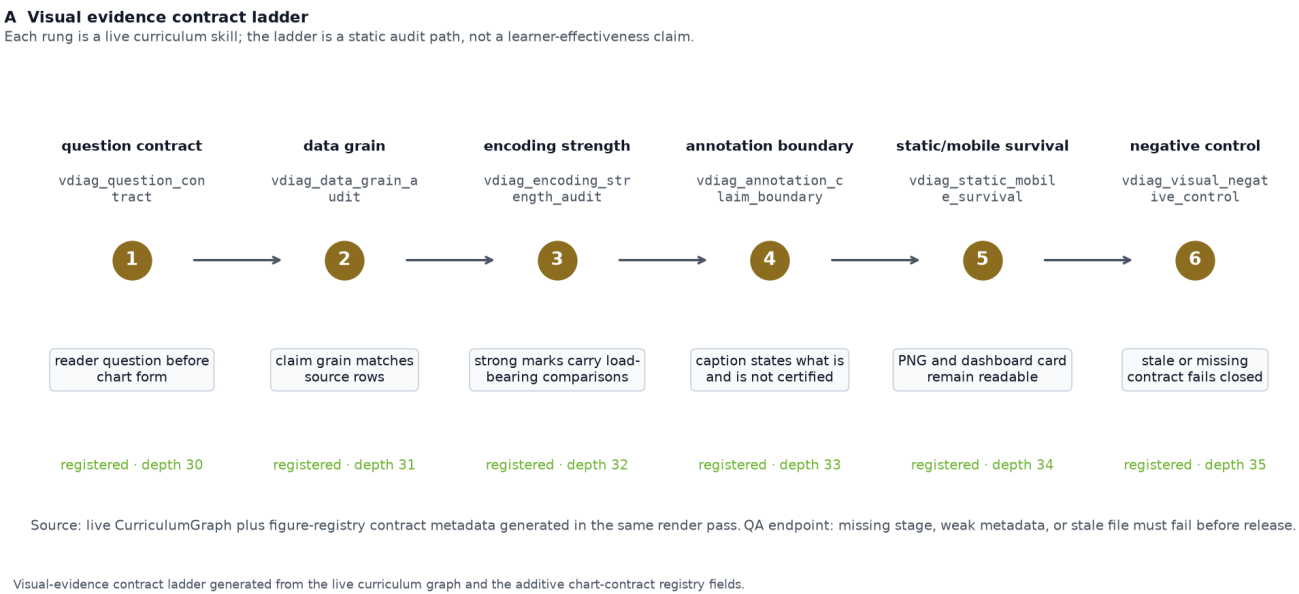


Figure 6: Visual-evidence contract ladder generated from the live curriculum graph and figure registry metadata. Six numbered rungs move from question contract through data grain, encoding strength, annotation boundary, static/mobile survival, and negative-control validation. Each rung names the skill identifier that carries the review obligation and reports whether the skill is registered in the graph. The figure documents the authoring and validation lifecycle for visual evidence; it does not claim learner effectiveness or external deployment performance.

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The scientific-claim layer adds one more audit view. fig. 7 is generated from manuscript `claim-support` anchors and the support matrix in `data/manuscript_claim_support.yaml`. Its rows separate computed, descriptive-count, simulation, literature, scope-limit, delivery and visual claims; its columns show whether each class is backed by claim tokens, manuscript variables, citations, figures, validators or limitation markers. This is not a statistical result. It is a release integrity check that prevents strong scientific wording from passing without a declared evidence class.

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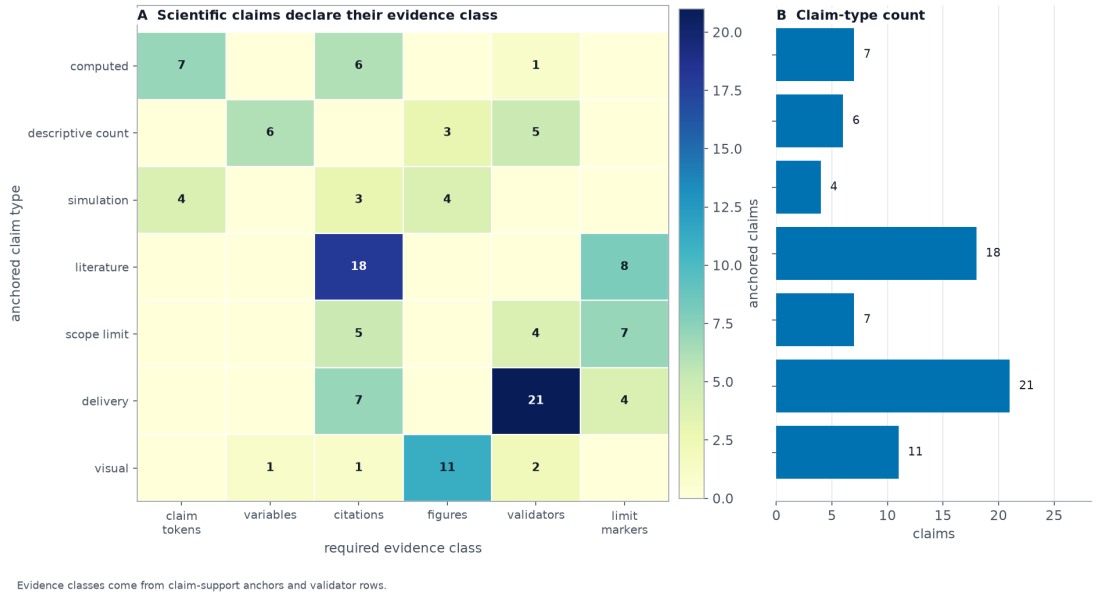


Figure 7: Scientific evidence matrix for manuscript claims generated from claim-support anchors and data/manuscript_claim_support.yaml. Rows are claim-support types, columns are evidence classes, cell counts show how many anchored claims use each evidence class, and the side panel counts claims per type. The purpose is epistemic classification rather than statistical inference: computed, descriptive, simulation, literature, scope-limit, delivery, and visual claims are forced to declare the kind of support they actually have.

The platform evidence above is intentionally separated from the small mathematical examples. Those examples are still central teaching artifacts, but their scientific meaning depends on the supplement's model assumptions and regime boundaries.